SAMADHAN AI

PROJECT PROGRESS REPORT

Overall Project Progress: 30%

Project Overview

Samadhan AI is an AI-powered civic complaint management platform designed to help citizens report civic issues and support authorities in analyzing, routing, tracking, and resolving complaints. The system combines AI-assisted analysis with backend validation, department routing, Service-Level Agreement (SLA) monitoring, escalation, and citizen verification. Human oversight remains part of consequential decisions, and AI recommendations do not independently authorize administrative actions.

Work Completed So Far — Samadhan AI

1. Project Planning and Requirement Analysis

- Finalized the problem statement and objectives of Samadhan AI as an AI-powered civic complaint management platform.
- Identified the key stakeholders: citizens, municipal officers, department administrators, and senior authorities.
- Defined the core complaint lifecycle, from complaint submission and analysis to routing, monitoring, escalation, and resolution verification.
- Identified the major functional requirements, including complaint classification, department routing, SLA monitoring, automated escalation, citizen notifications, and human oversight.
- Established the MVP scope and distinguished core requirements from future enhancements.

2. System Architecture and Technical Design

- Designed the overall system architecture, defining the responsibilities of the frontend, backend APIs, AI orchestration, agents, database, and supporting services.
- Selected a modular backend architecture in which the backend remains responsible for authorization, business rules, data validation, and state transitions.
- Defined the three core AI agents:
    1. Complaint Analyst Agent: Analyzes complaint information and generates structured analysis.
    2. Resolution & Routing Agent: Supports department routing, officer assignment, and handling of composite complaints.
    3. Monitoring & Escalation Agent: Supports SLA-breach handling and escalation workflows.
- Established the intended coordination approach using LangGraph.
- Documented the distinction between AI recommendations and authorized backend decisions.

3. Technology Stack Selection

- Selected Python and FastAPI for backend development.
- Selected PostgreSQL as the primary relational database and pgvector for vector-based retrieval.
- Identified Pydantic for request validation and structured AI outputs.
- Planned SQLAlchemy and Alembic for database access and schema migrations.
- Selected React and JavaScript for frontend development.

- Identified LangGraph, language/vision models, embeddings, and Retrieval-Augmented Generation (RAG) for the AI components.
- Defined the planned use of PostgreSQL-backed durable jobs for asynchronous processing, retries, and recovery.
- Identified pytest, OpenAPI, Postman, Git, and GitHub for testing, API documentation, version control, and collaboration.

4. Engineering Documentation

- Prepared the backend architecture documentation.
- Created the logical database model and Entity-Relationship (ER) diagram.
- Documented the complaint lifecycle and associated state transitions.
- Defined the intended user roles and authorization responsibilities.
- Documented parent-child complaint handling and independent SLA tracking for child complaints.
- Defined the intended resolution-evidence and citizen-verification workflows.
- Documented business events, background jobs, retry handling, idempotency, notification attempts, and audit logging.
- Conducted cross-document consistency reviews to align requirements, architecture, database relationships, and lifecycle behavior.
- Established the documentation baseline to guide implementation of the MVP.

5. Database Design and Creation

- Designed the database structure according to the project's documented requirements and logical data model.
- Created the database according to the designed schema.
- Established the database foundation for complaint records and subsequent backend workflows.
- Prepared the model to support the planned complaint lifecycle, assignments, SLA tracking, events, notifications, resolution evidence, and review workflows.

6. Initial Frontend Development

- Created the basic frontend foundation for Samadhan AI.
- Established the starting point for developing the platform's user interfaces.
- Prepared the frontend for subsequent integration with backend APIs and database-backed workflows.
- Established the foundation for the planned citizen, officer, and administrator interfaces.

7. First AI Agent — Complaint Analyst

- Implemented the first AI agent in the backend: the Complaint Analyst Agent.
- Established its role as the component responsible for understanding incoming complaint information and generating structured analysis.
- Defined its intended responsibilities, including:
    - Analyzing complaint descriptions and available image information.
    - Generating or refining complaint descriptions.
    - Suggesting complaint categories and priority levels.
    - Identifying missing information and representing uncertainty.
    - Producing structured outputs for downstream backend processing.
- Established the architectural principle that the backend must validate agent outputs before using them in consequential workflows.

8. Reliability, Security, and Workflow Design

- Defined the intended approach for deterministic SLA monitoring instead of repeatedly invoking an LLM to check deadlines.
- Established PostgreSQL-backed durable jobs as the planned foundation for asynchronous processing and retry handling.
- Documented idempotency requirements to reduce duplicate event processing and side effects.
- Defined the requirement for role-based authorization and backend-enforced state transitions.
- Documented auditability for privileged actions and exceptional administrative decisions.
- Established the intended citizen-verification workflow, including the 48-hour response window, reminder, and overdue-review process.

9. Overall Progress and Development Foundation

- Completed the initial project planning and core engineering documentation.
- Created the database according to the designed schema.
- Developed the basic frontend foundation.
- Implemented the first AI agent, Complaint Analyst.
- Established the architecture and technical baseline for developing the remaining agents and integrating the complete system.

Current Progress

The project is estimated to be 30% complete. This stage includes requirements and architecture documentation, database creation according to the schema, development of the basic frontend, and implementation of the first AI agent. These components establish the foundation for the next development phase. The 30% figure represents the team's overall project progress estimate.

Remaining Work and Next Steps

- Complete and integrate the Resolution & Routing and Monitoring & Escalation agents.
- Connect the frontend with backend APIs and database-backed workflows.
- Develop the complete complaint lifecycle, including image-first intake, location capture, department/officer assignment, and parent-child complaint handling.
- Implement deterministic SLA monitoring, durable background processing, notifications, escalation, and citizen verification.
- Strengthen authentication, authorization, evidence handling, auditability, and failure recovery.
- Perform unit, integration, security, and end-to-end testing, followed by demonstration deployment.

Conclusion

Samadhan AI has progressed beyond planning into initial development. The basic frontend, database created from the designed schema, and implemented Complaint Analyst Agent provide the foundation for continued implementation. The next phase will focus on completing the remaining agents, integrating the platform components, validating the end-to-end complaint workflow, and preparing a reliable demonstration.